

Why Blockchain-Based Digital Assets Are Owned on Decentralized Metaverse Platforms?

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Abstract

This study investigates ownership of blockchain-based digital assets in a decentralized metaverse platform enabled by Web3 and built on blockchain technology. Using the dual-process model of psychological ownership, the research examines factors influencing individuals' desire for ownership in this context. Through fsQCA analysis, necessary conditions and configurations driving ownership patterns are identified. Findings reveal a substitutive relationship between Transfer rights and Voting rights, as well as between Control usage and Voting rights. Additionally, a complementary relationship is observed between the presence of Self-investment and absence of Trade as core conditions within configurations. Notably, in the absence of Profits and Trade, possessing Transfer rights or Voting rights plays a crucial role in driving individuals to acquire higher levels of blockchain-based digital asset ownership. High profits are also highlighted as a motivator for digital asset ownership. These findings shed light on the psychological dynamics of ownership in decentralized metaverse platforms and DAOs.

Keywords: Blockchain-based digital assets; Web3 enabled decentralized metaverse platforms; decentralized autonomous organizations (DAOs); non-fungible tokens (NFTs); psychological ownership

1. Introduction

Web3 is the third generation of the internet, built on decentralized technologies like blockchain and decentralized autonomous organizations (DAOs), that aims to create a more democratized and secure network (Nabben, 2023). Web 3 applications could use non-fungible tokens (NFTs), augmented reality (AR), virtual reality (VR), and other technologies to facilitate the metaverse (Minevich, 2022). A Web3 enabled decentralized metaverse platform is a virtual world where the ownership and control of assets and resources are distributed among users, operating on a blockchain-

based, decentralized economic model (Blocktech Brew, 2023).

NFTs are a type of cryptocurrency and the concept of NFTs originally stems from a token standard of Ethereum (Wang et al., 2021). NFTs differ from the traditional cryptocurrencies because NFTs cannot be exchange equivalently, making it non-fungible and thus can be used for unique identification (Wang et al., 2021). NFTs are therefore connected with blockchain-based digital assets.

The metaverse and decentralized applications have been drastically stimulated accordingly. Decentralized metaverse platforms developed virtual worlds that use cryptocurrencies to buy blockchain-based digital assets and NFTs that guarantee complete ownership of avatars and virtual lands. The ownership of digital assets not only gives users the psychological ownership, but also the voting rights in the decentralized autonomous organizations (DAOs).

Prior research has explored various dimensions of digital assets and the metaverse, including conceptual examinations of the relationship between the two (e.g., Wang et al., 2021; Truong et al., 2023), and investigations into consumer behavior and brand engagement with NFTs in metaverse environments (e.g., Lee et al., 2023; Sung et al., 2023). However, empirical inquiries examining blockchain-based digital assets within a specific Web3-enabled, decentralized metaverse platform remain scarce, particularly regarding motivations for ownership. This represents a notable gap, as blockchain-based assets are core to decentralized metaverses, but factors driving user acquisition of such assets are not yet well understood.

Our study aims to address this gap through the theoretical lenses of psychological ownership and the dual process model (Morewedge, 2021) because they allow examination of both the implicit feelings and explicit judgments that comprise users' sense of digital asset ownership within the metaverse (Belk, 2014; Pierce et al., 2001). This study contributes to the theory on psychological ownership of digital assets by applying the dual-process model in the context of blockchain-based assets. It empirically distinguishes between and

establishes the relationship between implicit and explicit psychological ownership in decentralized metaverse settings. The research also holds practical implications for stakeholders involved with blockchain assets and metaverse platforms to enhance users' sense of ownership and increase engagement.

2. Theoretical background

2.1. NFTs and blockchain-based digital assets

Brown Sr et al. (2022) found that blockchain-based digital assets use cryptocurrencies, which preserve values similar as traditional currencies and tokens, to have unique characteristics on distributed platforms. Blockchain facilitates an ideal environment for decentralized virtual world and with NFTs. Belk et al. (2022) stated that in the case of digital assets, the need for authenticity validation from experts and agents can be largely reduced because the authenticity is secured by the blockchain, and the middleman can be removed between the producers and consumers through the utilization of blockchain as well.

NFTs have been investigated by extent studies from various perspectives with blockchain, to name a few, Schwiderowski et al. (2023) focused on the value creation and capture for NFTs as digital assets. Sung et al. (2023) examined consumer behavior regarding blockchain-based NFTs for luxury items in the metaverse. Naik et al. (2023) introduced a safe exchange for the exchange of NFTs for digital assets. Darshan et al. (2023) further proposed a framework for lending digital assets by using NFT and blockchain.

2.2. Blockchain-based digital assets and the Web3 enabled decentralized metaverse platforms

The metaverse refers to an interconnected and immersive digital environment, built upon reality-enhancing technologies, where users engage in a range of social, economic, and cultural activities through personalized avatars (Sung et al., 2023). As stated by the Software Policy and Research Institute (2021), the metaverse platform distinctly embodies characteristics such as providing a user-friendly virtual interaction space, leveraging advanced technologies to bridge the gap between virtual and actual reality, and establishing a novel economic market, collectively shaping an immersive and interconnected digital realm.

The emergence of the metaverse represents a shift from centralized platforms to user-owned environments enabled by blockchain technology and NFTs (Christodoulou et al., 2022). Decentralization through

blockchain allows true digital asset ownership, creates transparent and persistent records of transactions and activities, and gives users more control over their data and identities in these virtual spaces (Lu, 2019; Truong et al., 2023). As the metaverse moves toward decentralized models powered by blockchain, NFTs can represent unique verifiable tokens for digital assets like avatars, clothing, art, and virtual land, traded on blockchain ledgers (Anjum & Rehmani, 2022).

Digital assets and NFTs have been investigated in the study of metaverse conceptually. To name a few, Wang et al. (2021) studied systematically on the NFT ecosystems. Truong et al. (2023) examined the role of blockchain in the metaverse by using the analysis of digital assets management. However, blockchain-based digital assets have rarely been investigated empirically on a specific Web3 enabled decentralized metaverse platform, which underscores the importance of this study.

2.3. Blockchain-based digital assets and psychological ownership

Psychological ownership is related to the “proprietary feelings” regarding possessions in a symbolic and experimental manner (Belk, 2014). Pierce et al. (2001) defined psychological ownership as a feeling of possessively experiencing an object, which can be material or immaterial, in other words, the object is considered as “MINE”. Prior research on psychological ownership has utilized qualitative approaches to examine organizational contexts (Pierce et al., 2001) and consumer behavior perspectives (Peck & Luangrath, 2023). Quantitative methods have also assessed psychological ownership in relation to consumer-brand relationships (Zhang et al., 2023). However, empirical examinations of psychological ownership remain scarce regarding digital assets, particularly within decentralized metaverse platforms. Despite the centrality of blockchain-based assets to decentralized virtual worlds, factors motivating users to acquire digital assets in these emerging environments are not yet well understood.

According to the dual-process model of psychological ownership proposed by Morewedge (2021), the psychological ownership can be split into implicit and explicit psychological ownership; the implicit associate which is created between an individual and an object arises to give the impression or an inference that the object is MINE, which can be considered as System 1 (i.e., *implicit psychological ownership*) (Morewedge, 2021). The impression or inference is accordingly accepted, rectified, or denied by the rule-oriented explicit judgments, which can be

referred to as System 2 (i.e., *explicit psychological ownership*) (Morewedge, 2021).

Both implicit and explicit psychological ownership are included in this dual-process model because they posit that both intuitive feelings of ownership and reasoned judgments jointly shape the subjective experience of objects being "MINE" (Morewedge, 2021). The implicit and explicit processes provide complementary routes through which perceptions of ownership manifest. This dual process perspective is relevant for understanding digital assets on decentralized metaverse platforms, as the platforms obfuscate traditional notions of ownership through community governance and transparency of transactions (Fan et al., 2023). This could enable divergences between implicit assumptions of ownership and explicit judgments.

3. Research methodology

3.1. Data collection

Decentraland was chosen as the research subject due to two reasons. To start with, Decentraland qualifies as a metaverse platform because it provides a persistent 3D virtual environment transcending any single game or experience, where users can freely explore, interact, and participate in diverse social activities through avatars (Decentraland, 2023). The second reason for selecting Decentraland as the research subject lies in its status as a representative metaverse platform that embodies the concept of "decentralization". This is evident through its governance by a DAO (Decentralized Autonomous Organization), which empowers every user within the community to participate and cast votes on proposed changes they would like to see adopted or rejected (Decentraland, 2023).

The research participants were approached in the Web3 enabled decentralized metaverse platform, Decentraland by the research team. Once the research participants gave their consent to participate, the survey questionnaire link was sent to them. An incentive (a gift card) was given to the research subjects once the survey was completed.

3.2. Measurements

The measurement scales are specified for implicit psychological ownership (i.e., self-investment, object knowledge, and self-congruity), explicit psychological ownership (i.e., control usage, profit, trade, transfer rights, and voting rights), and the dependent variable of ownership.

The constructs are chosen as they are from the dual-process model, the implicit psychological ownership factors have been shown to create intuitive associations between one's self-concept and the target object, giving rise to a sense of possession. In contrast, the factors in explicit psychological ownership are based on the legal rights of control over usage, profit generation, trading, transferring, and voting, as conscious consideration of these rights requires deliberative judgments about whether legal ownership is truly held (Morewedge, 2021).

The measurement items are based on and adapted from the previous literature, so that the validity and reliability of the research items are guaranteed, and our research aim can be better described and explained.

Table 1 provides the operational definitions of variables that were employed in this research. Decentraland users were asked to answer the questionnaires. We employed a seven-point Likert scale to examine how Decentraland users feel from strongly disagree to strongly agree towards the implicit psychological ownership since it is a subjective perception from users. On the other hand, Decentraland users were required to write down their objective responses for the items in explicit psychological ownership, such as the question of "how much profit you have earned in buying blockchain-based digital assets per month?" for the item of profit.

Table 1. Operational definitions of constructs

Construct	Operational Definition	References
<i>Implicit Psychological Ownership</i>		
Self-investment	The extent to which individuals feel they have invested efforts or labor in blockchain-based digital assets.	Bagga et al. (2019); Nancekivell et al. (2019)
Object knowledge	The extent to which individuals have related knowledge about blockchain-based digital assets.	Jarvenpaa & Staples (2001)
Self-congruity	The extent to which individuals share meaningful associations between blockchain-based digital assets and themselves.	Morewedge (2021)
<i>Explicit Psychological Ownership</i>		
Control usage	The extent to which individuals legally own and protect their	Lazaro & Metayer (2015)

	blockchain-based digital assets from unwanted uses.	
Profit	The extent of the difference between the received revenues and the incurred costs in buying blockchain-based digital assets.	Valentini (2013)
Trade	The extent to which individuals buy and sell blockchain-based digital assets.	Heflin & Shaw (2000)
Transfer rights	The extent to which individuals are allowed to transfer contractual rights and responsibilities to another party.	Ramadhani (2020)
Voting rights	The extent of the ability of participants or token holders within the DAO to participate in decision-making processes by casting votes on various matters (such as governance issues or protocol upgrades).	Sims (2019)
Ownership	The extent to which individuals own blockchain-based digital assets on the blockchain.	Belk et al. (2022)

3.3. QCA

In this study, we utilized fsQCA (small-N QCA) as a research method to explore the factors influencing the ownership of blockchain-based digital assets on a decentralized metaverse platform. The reasons behind people's desire to own blockchain-based digital assets are multifaceted and complex. The high ownership of these assets is not solely attributed to a single factor, but rather results from a combination of numerous factors. Conventional regression methods can only determine the net effect of individual factors on the outcome, failing to capture the intricate causal relationships between variables (Rihoux & Ragin, 2009). Consequently, they fall short in fully explaining the occurrence of specific outcomes such as high ownership of digital assets.

To address these limitations, this study employs the Qualitative Comparative Analysis (QCA) method, which is rooted in set theory configuration analysis based on Boolean algebra (Ragin, 2000, 2009). By examining the sufficient and necessary subset relationships between antecedents and outcomes, QCA allows for the exploration of how complex social issues emerge as a result of multiple simultaneous causes and effects. This approach proves effective in overcoming the aforementioned challenges (Fiss, 2007). Therefore, this paper utilized the QCA method to investigate the precise pathways that lead individuals to express a sense of ownership towards blockchain-based digital assets.

4. Results

In general, the research process involves four essential steps: calibration, case categorization, truth table construction, and counterfactual analysis. To carry out our analysis, we employed the fsQCA software, which facilitated the execution of these steps.

4.1. Calibration

The calibration process involves transforming the original measures into fuzzy-set membership scores, which indicate the extent to which they belong to a target set (refer to Table 2). These fuzzy-set membership scores are continuous and range from 0 to 1. During the calibration process, we identified three qualitative breakpoints for the original measures, namely Ownership, Self-investment, Object knowledge, and Self-congruity, based on our research data: (1) the threshold for full membership (fuzzy-set score = 0.95), (2) the cross-over point (fuzzy-set score = 0.50), and (3) the threshold for full non-membership (fuzzy-set score = 0.05).

However, we observed that the data distributions for both Profit and Trade were biased, with a significant majority of values equal to 1. Consequently, for Profit (Trade), we set the value to 0 only when the original value was 1; otherwise, it remained as 1.

Table 2. Calibration

Determinants	Fuzzy Set	Calibration Standard	Threshold Value at		
			0.95	0.5	0.05
			Full Membership	Cross over Point	Full Non-membership
Ownership	high	Research data	6.65	4.00	1.00

Self-investment	high	Research data	6.75	3.75	1.59
Object knowledge	high	Research data	6.21	4.00	2.00
Self-congruity	high	Research data	6.33	4.25	2.00

4.2. Case categorization

After completing the calibration process, we proceeded with the calculation of membership scores for each case across all possible configurations. Based on these scores, we were able to assign each case to the specific configuration that had the highest membership score.

4.3. Truth table construction

Upon completing the calibration and case categorization, we constructed the truth table. Generally, a minimum frequency threshold of 1 is commonly employed to exclude configurations that have not been observed, while a higher threshold is recommended to exclude configurations that may appear insignificant and prevent the generation of excessively complex results in large samples with numerous conditions (Douglas et al., 2020). However, given our small sample size ($n=46$), we included all observed configurations in our analysis. Thus, for this study, the frequency cutoff was set at 1. Following Marx's (2010) guidance, given our dataset of 46 entries, we have adjusted the number of conditions to eight, aligning with the specifics of our study.

In line with Ragin (2009), we established the threshold for raw consistency at 0.75, and the PRI consistency threshold at 0.6.

4.4. Counter-factual analysis

The final solutions were derived by conducting a counterfactual analysis. However, due to the presence of unobserved instances resulting from various combinations of conditions, a number of remainders were generated. In the process of counterfactual analysis, we determined whether these remainders should be included in the final solutions based on established empirical evidence or commonly accepted knowledge.

5. Discussions

5.1. Analysis of necessary conditions

To ensure accurate fuzzy set configuration analysis, researchers should initially perform a necessary conditions analysis as a fundamental step (Fiss, 2011). This entails examining whether any conditions are necessary for the results before proceeding with the standard analysis. By undertaking this critical step, researchers can establish the foundation of their analysis on accurate data and prevent potential errors that may arise from disregarding necessary conditions. Disregarding the analysis of necessary conditions can result in the omission of necessary conditions in the parsimonious solution, thereby impacting the identification of core conditions and conditional configurations.

Generally, consistency scores greater than 0.9 are considered as the threshold for a necessary condition (Douglas et al., 2020). The results of the necessity analysis of the antecedents of ownership are presented in Table 3. The highest consistency observed among all antecedents is 0.873, which is slightly below the threshold. However, the identification of necessary conditions is a complex undertaking. In fsQCA, the analysis of necessity is based on the notion of 90% supersets (i.e., a necessary consistency greater than 0.9) and relies on empirical judgments (Ragin, 2009). It is important to note that there may exist instances where "opposite logic" cases occur, meaning conditions with a consistency lower than 90% could also be deemed necessary conditions. Consequently, recent research tends to adopt a default stance of retaining necessary conditions.

The purpose of necessary condition analysis is to prevent the omission of core conditions in configurational analysis (Ragin, 2009). Conditions with a consistency value exceeding 90% in the necessary conditions analysis are regarded as necessary conditions. Importantly, these necessary conditions always hold the status of core conditions, regardless of whether they are explicitly recognized as such in the configurational analysis. Therefore, it is imperative to recognize and consider the identified necessary conditions as core conditions within the context of this study.

Table 3. Analysis of necessary conditions for high ownership

Sets of conditions	Consistency	Coverage
Self-investment	0.860	0.751
~Self-investment	0.588	0.430

Object knowledge	0.873	0.712
~Object knowledge	0.572	0.445
Self-congruity	0.873	0.679
~Self-congruity	0.602	0.492
Control usage	0.531	0.486
~Control usage	0.469	0.330
Profit	0.127	0.773
~Profit	0.873	0.372
Trade	0.203	0.530
~Trade	0.797	0.374
Transfer rights	0.463	0.499
~Transfer rights	0.537	0.339
Voting Rights	0.441	0.539
~Voting Rights	0.559	0.330

Note: ~ means the absence of. For example: ~ Profit = absence of high Profit.

5.2. Analysis of configurations for high ownership

The analysis of configurations aims to examine whether a set of multiple conditions forming a configuration is a subset of the result set, thereby indicating the sufficiency of the configuration to achieve the desired outcome. Utilizing the truth table, we employed standard analysis to derive complex solutions (without considering logical remainders), parsimonious solutions (incorporating all logical remainders), and intermediate solutions (utilizing reasonable logical remainders). Typically, intermediate solutions are utilized to identify the configurations and are reported in the study. These solutions are combined with the parsimonious solutions to distinguish the core and peripheral conditions (Ragin, 2009). The outcomes of the standard analysis are outlined in Table 4.

The high overall solution consistency of 0.900 suggests that the set-subset connections identified in Table 4 are robust and well-supported by the data based on the instructions of Greckhamer et al. (2013). However, the overall solution coverage is 0.379, indicating that while the relationships discovered are consistent, they apply to a limited number of respondents. This coverage, being lower than perfect coverage, also implies that there might be other causal conditions leading to high ownership in blockchain-based digital assets (Wagemann et al., 2016).

5.2.1. Basic results of four configurations. The four configurations delineate distinct pathways that lead to high ownership of digital assets. Configuration 1 identifies Self-investment, Transfer rights, and the absence of Trade as core conditions, accompanied by a

peripheral emphasis on Control usage and the absence of Profit. This indicates that individuals who prioritize Self-investment and possess Transfer rights are likely to exhibit substantial ownership of digital assets, even without engaging in frequent trades or generating significant profits.

In Configuration 2, Self-investment, Voting Rights, and the absence of Trade are identified as core conditions, with Control usage and the absence of Profit serving as peripheral conditions. This suggests that individuals who prioritize Self-investment and possess Voting rights are likely to demonstrate substantial ownership of digital assets, irrespective of their involvement in infrequent trades or lower profit levels.

Similarly, Configuration 3 highlights Self-investment, Transfer Rights, and the absence of Trade as core conditions, with Voting rights and the absence of Profit considered as peripheral conditions. This configuration also indicates that individuals who prioritize Self-investment and possess Transfer rights are likely to possess significant ownership of digital assets, even if they engage in fewer trades or experience lower profit levels. Our findings are consistent with Kamleitner & Mitchell (2018) in terms of self-investment, and Toygar et al. (2013) in terms of transfer rights.

Lastly, Configuration 4 identifies the presence of Profit as the sole core condition, implying that high profitability serves as a key determinant for individuals holding substantial levels of digital asset ownership, which is consistent with Restuputri et al. (2023).

5.2.2. Substitutive relationships. A comparison between Configurations 1 and 2 uncovers a substitutive relationship between Transfer rights and Voting rights, as both conditions emerge as fundamental factors in their respective configurations. Configurations 1 and 2 demonstrate that the presence of either transfer rights or voting rights serves as a core condition that can substitute the other in determining digital asset ownership. This implies that possessing either transfer rights or voting rights can facilitate higher levels of digital asset ownership in the absence of profits and trade. By establishing this substitutive relationship between these two factors, valuable insights can be gained into the interplay and influence of different elements on patterns of blockchain-based digital asset ownership.

It is found by Perlman (2019) in a crypto-asset regulatory framework that both transfer rights and voting rights are vital factors in crypto assets tokens. The possible explanation of our finding could be that Transfer rights and Voting rights may serve overlapping functions in terms of asserting control and influence over digital assets. Individuals with Transfer rights can exercise control by transferring ownership or making

decisions related to the assets, while individuals with Voting rights can participate in decision-making processes that affect asset ownership and management within the platform or ecosystem of a DAO (Liu, 2023). Thus, the presence of one of these rights may compensate for the absence of the other, leading to similar levels of digital asset ownership.

Likewise, Configurations 1 and 3 indicate a substitutive relationship between Control usage and Voting rights, which are categorized as peripheral conditions. These findings suggest that the presence of Control usage and Voting rights can act as interchangeable factors in influencing digital asset ownership. This implies that individuals who possess a higher degree of Control usage over their assets may require fewer Voting rights, and vice versa, to achieve similar levels of ownership.

A possible explanation might be that individuals with a higher degree of Control usage over their digital assets may feel a greater sense of ownership and control, which reduces their reliance on Voting rights. They may perceive their ability to manage and protect their assets as sufficient without the need for voting. Having more autonomy in utilizing one's assets on a day-to-day basis can provide a perception of control and possession, thus compensating for a lack of formal Voting rights.

Another plausible explanation is that the substitutive relationship may be influenced by the nature of the blockchain-based digital asset itself. Certain types of assets, such as NFTs, may have unique characteristics that affect the importance of Control usage and Voting rights.

5.2.3. Complementary relationships. A comparison of Configurations 1, 2, and 3 highlights a complementary relationship between the presence of Self-investment and the absence of Trade as core conditions within the configurations. This indicates that Self-investment and the absence of Trade, identified as core factors in Configurations 1, 2 and 3, mutually reinforce each other in influencing blockchain-based digital asset ownership. The inclusion of Self-investment and the exclusion of Trade in these configurations suggest that individuals who engage in self-investment, such as actively managing and growing their digital assets, are more likely to prefer an environment without trade-related activities. Conversely, the absence of trade-oriented transactions may promote a greater focus on self-investment strategies. This complementary relationship underscores the interplay between these factors and their combined influence on blockchain-based digital asset ownership patterns.

5.2.4. Supplementary results. Notably, in the absence of Profits and Trade, possessing Transfer rights or Voting rights appears to play a crucial role in driving individuals to acquire higher levels of blockchain-based

digital asset ownership. These conditions provide individuals with control, decision-making power, and a sense of ownership, which can compensate for the lack of substantial profits or frequent trading activity.

Moreover, Configuration 4 underscores the significance of high profits as a strong motivating factor for individuals to hold substantial amounts of digital assets. This finding aligns with Bull & Harttraft (2020) and Sazandrishvili (2020) with emphasis on the importance of profits in terms of cryptocurrency and blockchain law and asset tokenization. Our results implied that when individuals perceive the potential for substantial financial gains, it acts as a driving force for them to accumulate and retain significant ownership of blockchain-based digital assets.

Table 4. Configurations for high ownership

Determinants	Configurations for high Ownership			
	C1	C2	C3	C4
Self-investment	●	●	●	●
Object knowledge	●	●	●	●
Self-congruity	●	●	●	●
Control usage	●	●		●
Profit	⊗	⊗	⊗	●
Trade	⊗	⊗	⊗	●
Transfer rights	●		●	●
Voting rights		●	●	●
Raw Coverage	0.110	0.130	0.167	0.093
Unique Coverage	0.050	0.070	0.106	0.093
Consistency	0.995	0.905	0.857	1.000
Overall Solution Coverage	0.379			
Overall Solution Consistency	0.900			

Note: ● presence of a core condition; ⊗ absence of a core condition; ● presence of a peripheral condition; ⊗ absence of a peripheral condition; blank spaces denote a 'don't care' condition

5.3. Theoretical implications

Based on the theory of blockchain-based digital assets and psychological ownership outlined in the study, the theoretical implications of the results and analysis can be considered in the following ways.

To start with, psychological ownership has been used to examine organizational and consumer behavior qualitatively (Peck & Luangrath, 2023; Pierce et al., 2001), as well as in consumer-brand relationships (Zhang et al., 2023). However, examinations of psychological ownership remain scarce regarding digital assets. This study thus is one of the first attempts to empirically investigate blockchain-based digital assets using a psychological ownership approach. To be specific, the findings highlight the unique nature of ownership in the context of blockchain-based digital assets. Unlike traditional physical goods, digital assets involve the reproduction and electronic transmission of the source copy (Stefik, 1997). This challenges the traditional notion of ownership and necessitates a deeper understanding of how individuals perceive and experience ownership in the digital realm. The theoretical implications of the study shed light on the evolving concept of ownership and its implications for the digital age.

Secondly, the study contributes to the theoretical understanding of the psychological ownership theory, particularly the dual process model. This model has been proposed by Morewedge (2021) theoretically and been used to investigate online game addiction (Wang et al., 2021). However, this study constitutes a novel exploration into empirically examining and differentiating between implicit and explicit psychological ownership in the novel context of blockchain-enabled virtual environments. Our study aims to enrich the understanding of this theoretical model as we believe both the implicit feelings and explicit judgments are vital in the ownership of digital assets.

Furthermore, the results suggest that conditions such as transfer rights, voting rights, and control usage play a significant role in influencing individuals' psychological ownership of digital assets. These conditions provide individuals with a sense of control, decision-making power, and autonomy over their assets, contributing to their psychological ownership experience (Dawkins et al., 2017). This theoretical implication emphasizes the importance of perceived control and rights in shaping individuals' psychological relationship with blockchain-based digital assets.

Lastly, we believe that this study contributes to the influence of profits on ownership. The finding that high profits serve as a motivating factor for individuals to hold substantial digital asset ownership has theoretical implications for the intersection of financial incentives and psychological ownership. It suggests that financial gains can influence individuals' attachment and possessive feelings towards digital assets (Belk et al., 2022), providing insights into the economic factors that drive ownership experiences.

By examining the theoretical implications of the results and analysis in the context of blockchain-based digital assets and psychological ownership, the study contributes to the broader theoretical understanding of ownership dynamics in the digital realm. It highlights the need for further research and theoretical development in exploring the complex interplay between technological advancements, psychological processes, and ownership experiences in the context of digital assets.

5.4. Practical implications

The findings of this study hold several practical implications for various stakeholders in the context of blockchain-based digital assets and decentralized metaverse platforms. To start with, this study provides practical implications for platform design and governance. The results highlight the importance of providing users with transfer rights and voting rights, which can fulfill ownership needs by enabling control and participation in decision-making. According to psychological ownership theory, control and ability to influence decisions are key antecedents that cultivate feelings of possession (Pierce et al., 2001). Platform developers and managers can focus on designing and implementing features that enable users to have control over their assets and participate in more decision-making processes on DAOs (Zhao et al., 2022). This can enhance user engagement, satisfaction, and trust in the platform, ultimately promoting higher levels of blockchain-based digital asset ownership.

Secondly, this study could contribute to investor education and engagement. The study suggests that promoting investor education regarding the benefits of self-investment and the significance of transfer rights and voting rights can encourage individuals to actively invest in blockchain-based digital assets. Educating investors about the potential returns and long-term value of digital assets, as well as their rights and participation opportunities, can enhance their willingness to acquire and hold digital assets (Hargrave et al., 2019).

Thirdly, we believe that this study can contribute to risk management and diversification. Understanding the factors that drive blockchain-based digital asset ownership, such as transfer rights, voting rights, and profitability, provides insight into user psychology and motivations. By considering how these antecedents shape ownership tendencies, investors and platform users can make informed decisions regarding risk management and portfolio diversification. Investors can consider the role of these conditions when constructing their investment strategies, aiming to balance risk and maximize potential returns. Balancing assets based on

these psychological drivers of ownership can assist in diversification and measured risk-taking.

5.5. Limitations and future research

While the present study provides valuable insights into the factors influencing ownership of blockchain-based digital assets and their implications for psychological ownership, it is important to acknowledge certain limitations and suggest directions for future research.

Firstly, this study utilized a relatively small sample size, which may limit the generalizability of the findings. Future studies could consider larger and more diverse samples to enhance the external validity of the results.

Secondly, the study focused on a specific context of decentralized metaverse platforms and blockchain-based digital assets. However, ownership dynamics may vary across different contexts and platforms. Future research could explore how contextual factors, such as different types of digital assets or alternative metaverse platforms, influence ownership experiences.

Additionally, the current study captured a snapshot of ownership patterns and their underlying conditions. Conducting longitudinal studies would enable researchers to examine ownership dynamics over time and investigate the causal relationships between the identified conditions and ownership outcomes.

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